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Research article

Performance Analysis of Decision Tree in Multimodal Feature Extraction on Retinal Fundus Image using Various Segmentation

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ABSTRACT

This research focuses on improving accuracy in diagnosing eye diseases through retinal image analysis. The main problem encountered is the uncertainty in the diverse fundus image dataset. Therefore, classification was performed using three different feature extraction methods, namely Haralick Texture Features, HuMoments, and Local Binary Pattern (LBP), which were evaluated in combination with three different segmentation techniques, namely Canny, Threshold Function, and Sobel. Utilizing the Kaggle dataset comprising multiple classes of retinal conditions, the study focused on six specific classes for training.The results revealed that Local Binary Pattern (LBP) as the standout performer, consistently surpassing HuMoments and Haralick across all segmentation techniques. LBP exhibited superior accuracy, precision, recall, and F1-score, showcasing its adaptability and generalizability to diverse retinal conditions. Noteworthy is LBP's sustained dominance even when paired with different segmentation methods, reaffirming its efficacy in capturing discriminative features critical for accurate disease classification. The practical implication is to improve the accuracy of diagnosing eye diseases through medical image analysis. This has an important impact on healthcare, enabling more timely treatment. Future research could explore integrating this method with deep learning and further evaluate its clinical impact on medical practice. Thus, this study provides a brief view of how to address uncertainty in fundus image datasets to improve eye disease diagnosis through more advanced medical image analysis.

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1. Introduction

The human eye is a marvel of biological engineering that allows us to see a dynamic world. However, many eye diseases threaten this precious sense, with retinal disorders being one of the causes. According to the World Health Organization (WHO), there are more than 253 million people in the world who suffer from blindness or visual impairment. Diagnosis and monitoring of retinal conditions play a crucial role in ophthalmology, aiming to prevent vision loss and provide protection through timely action. Over time, medical imaging technology has become an essential foundation in such endeavors [1],[2]. Breakthroughs in digital imagery have brought new opportunities for more accurate analysis and diagnosis in this domain.

In this context, the utilization of multimodal feature extraction techniques coupled with advanced segmentation methods presents a promising avenue for improving the accuracy and efficiency of

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disease identification. This research delves into the performance analysis of decision trees [3] within the framework of multimodal feature extraction, leveraging three distinct segmentation techniques—Canny, Threshold, and Sobel has emerged as a promising approach to improve accuracy rates in the classification and early detection of potentially harmful eye diseases.

A significant challenge in retinal image analysis is the diversity of complex details contained in these images [4]. Therefore, accurate and representative feature extraction is required to enable proper disease classification. This study addresses the need for robust methods by focusing on specific diseases such as Dry AMD, Wet AMD, Mild DR, Moderate DR, Severe DR, and Cataract, aiming to enhance the precision of diagnosis through the integration of diverse segmentation and feature extraction approaches [5].

The main objective of this study was to assess the performance of decision trees in the context of retinal fundus image analysis on various retinal conditions based on features obtained with a multimodal feature extraction framework. Specific objectives include evaluating the impact of various segmentation techniques to improve the quality and distinctiveness of features obtained from the images. While the use of feature extraction namely, HuMoments, Haralick and Local Binary Pattern (LBP) tests on the overall classification accuracy in the depiction of features related to shape, gradient, and texture. This study utilizes a dataset sourced from Kaggle, focusing on the training set and narrowing down the analysis to six specific classes. The limitations include the scope of diseases covered and potential variations in image quality. Future studies could explore expanding the dataset size and incorporating additional retinal diseases for broader applicability.

Previous studies have shown that retinal fundus images are important for early detection of eye problems. With the advancement of technology, Convolutional Neural Network (CNN) has been widely used to improve the accuracy of image segmentation and classification. Several studies reveal that CNNs are effective in reducing reliance on human experts, although there are still some limitations in their application [6]. The research [7] also shows that Diabetic Retinopathy (DR) can be detected automatically through feature extraction with Convolutional Neural Networks (CNN), which provides high accuracy in retinal image classification. The latest CNN model tested on IDRiD, MESSIDOR, and KAGGLE datasets shows superior performance with up to 99.89% accuracy on binary classification and 99.59% on multiclass, especially when combined with the J48 classifier. The research [8] also discusses the role of the retina and the high cost of eye examinations. CNN technology is proposed for disease detection. This study uses CNN-based SVM and DenseNet to detect Drusens, Optic Disc Cupping, and Tessellation in 534 fundus images, achieving an accuracy of 93.21% with DenseNet. in a study [9] using ResNet50V2 and InceptionV3 to classify DR in fundus images, comparing three classes: Normal, Mild/Moderate DR, and Severe DR. ResNet50V2 achieved better results with 80% accuracy, 80% sensitivity, 92% specificity, and faster processing at 12.42 seconds per step. Similarly, this study [10] develops a deep learning-based DR detection system using Optimized Iterative Thresholding (O-IT) for blood vessel segmentation. A hybrid Shark Smell-Jaya Optimization (SS-JO) algorithm is employed to enhance segmentation, replacing CNN with an optimized Long Short-Term Memory (LSTM) framework. The SS-JO-CN-LSTM model demonstrates accuracy improvements of 44%, 29%, 19%, 6%, 4%, and 15% over SVM, NN, CNN, LSTM, CN-LSTM, and FR-CSA-NN+CNN.

This study expected to serve as a basis for medical professionals and researchers to improve accuracy in the diagnosis of eye diseases, which in turn can aid in more informed medical decision-making. Furthermore, by creating a better understanding of the synergistic role between segmentation and multimodal feature extraction, this research has the potential to guide the development of more advanced and reliable medical image analysis methods in the future.

2. Materials and Methods

2.1. Materials

The dataset for this research is acquired from Kaggle, specifically the repository available at [www.kaggle.com/datasets/kssanjaynithish03/retinal-fundus-images]. The dataset comprises three subsets: training (train), testing (test), and validation (val). For the purposes of this study, emphasis is placed on the training set, which includes instances representing six distinct classes: Dry AMD, Wet AMD, Mild DR, Moderate DR, Severe DR, and Cataract. The selection of these classes aligns with the research focus on specific eye diseases. An example dataset can be seen in Figure 1.

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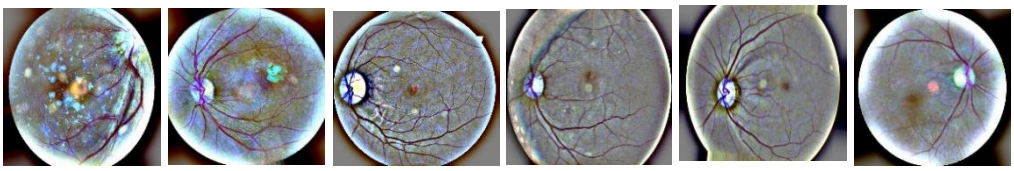

(a) (b) (c) (d) (e) (f)

Fig. 1. Example of Dataset: (a) Dry AMD; (b) Wet AMD (c) Mild DR (d) Moderate DR (e) Severe DR (Cataract)

The data collection process involves preprocessing the retinal fundus images by applying the selected segmentation method and then extracting relevant features. The curated dataset was then divided into training and testing sets, to ensure the robustness of the model evaluation. Table 1 shows general information on the dataset used in this study.

Table 1. Dataset Information

Dataset	Name of class	Number of Cases
Retinal Fundus Images	Dry AMD	1276
	Wet AMD	545
	Mild DR	2294
	Moderate DR	4982
	Severe DR	1635
	Cataract	1369

This dataset consists of a collection of retinal fundus images from various sources, totaling 20,622. The dataset includes several classes, each representing a different retinal condition, such as Dry AMD, Wet AMD, Mild DR, Moderate DR, Severe DR, and Cataract. This dataset was chosen as it provides a representative and diverse set of retinal fundus images relevant to the research objectives. The preprocessing step involved resizing all retinal fundus images [3],[11].

2.2. Methods

2.2.1. Research Flow

The research design consisted of 5 main stages that were well organized with aspects illustrated in Figure 2.

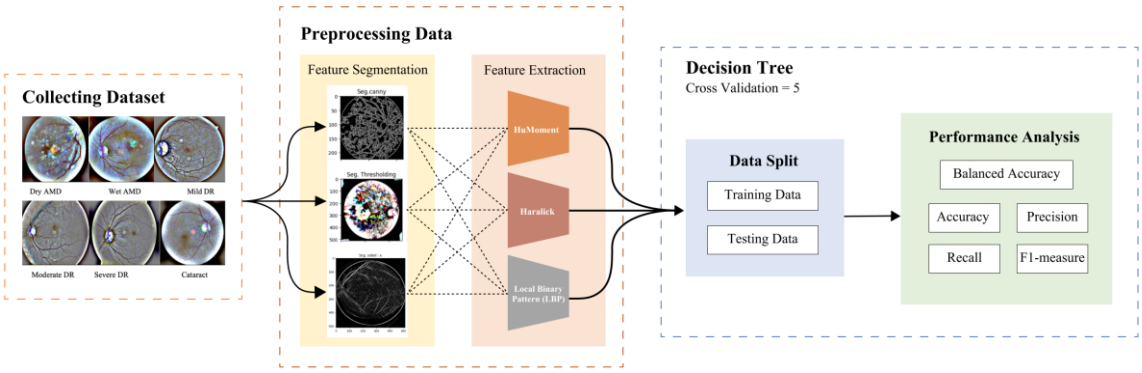

Fig. 2. Research Design

2.2.2. Preprocessing

Data preprocessing is a very important early stage in image analysis. Preprocessing is important because it enables noise removal and image quality enhancement, making the resulting image more suitable for the next stage of analysis [12]. In this study, data preprocessing involves two main stages, namely image segmentation and feature extraction.

In segmentation, the first stage in data analysis is the application of image segmentation using the Canny method. Canny edge detection is a reliable technique for identifying edges and discontinuities in images with a multi-stage algorithm for its ability to effectively identify edges in an image while suppressing noise. [13]. In this study, Canny edge detection was applied to pre-processed retinal fundus images. This technique highlights the boundaries of the image, emphasizing sudden intensity changes that indicate structural features within the retina such as blood vessels or lesions [14].

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This process is used to extract edge-based information from retinal images, which is important for distinguishing retinal conditions [15]. This aids subsequent analysis by providing a clear representation of the structural features. The mathematical formula underlying this method in 1,

$$E(x,y) = \sqrt{G_x(x,y)^2 + G_y(x,y)^2} \quad (1)$$

After obtaining the gradient components on the x and y axes, the gradient magnitude is calculated by taking the square root of the sum of their squares. The x gradient filter measures the intensity change between two adjacent pixels in the horizontal direction, while the y gradient filter measures the intensity change between two adjacent pixels in the vertical direction. This gives a magnitude value at each point in the image, indicating the extent of intensity change. In Figure 3, the results of image segmentation using Canny features on the dataset are shown.

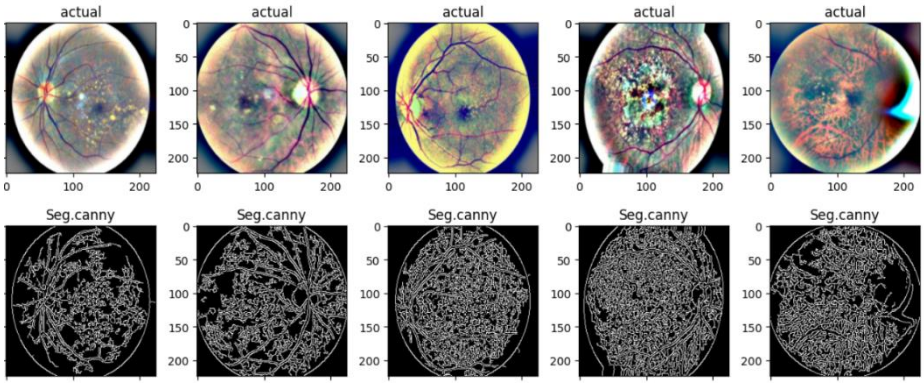

Fig. 3. Visualization of Canny Feature Segmentation Results

The second feature segmentation used is Thresholding. Thresholding is a simple technique used to classify pixels in an image based on their intensity against a certain threshold value. After the initial segmentation process using the Canny method, thresholding is used to smooth and separate certain regions in the segmented image [16]. By determining an appropriate threshold, this technique allows for increased visibility of certain structures or anomalies in the retina. The purpose of thresholding segmentation is to improve the segmentation results initially produced by the Canny method [17]. By adjusting the threshold, the segmentation process can be customized to highlight special features, which then helps in subsequent feature extraction and analysis. The mathematical formula underlying this method is equation 2,

$$I(x,y) = \{\, 1, \text{if I(x,y)} > T, 0, if\ I(x,y) \leq T\} \quad (2)$$

$I(x,y)$ is the intensity of the pixel at coordinate (x,y) in the image, and T is the specified threshold value. When the pixel intensity ($I(x,y)$) exceeds the threshold value (T), the pixel will be classified as a pixel with a value of 1 (white), whereas if the intensity is less than or equal to the threshold value, the pixel will be classified as a pixel with a value of 0 (black). In Figure 4, the results of image segmentation using tresholding features on the dataset are shown.

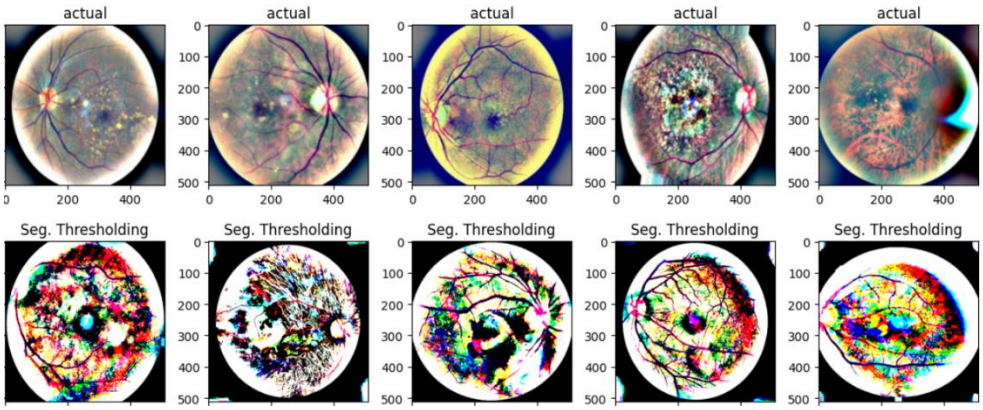

Fig. 4. Visualization of Thresholding Feature Segmentation Results

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The last segmentation used is Sobel. This segmentation method is based on gradients that highlight areas with sharp intensity changes in the image [18]. It involves convolving the image with a Sobel kernel to calculate gradients in the horizontal and vertical directions [19]. This allows the operator to capture directional information about the edges, making it possible to identify not only the presence of an edge but also its orientation. Applied to retinal fundus images, it highlights details such as blood vessels. This method is specialized for capturing gradient-based features in retinal images, aiding the extraction of important information for diagnosis [20]. The equation of sobel is in 3,

$$E = \sqrt{Gx^2 + Gy^2} \tag{3}$$

This results in a gradient image where the pixel intensity describes the extent to which intensity changes occur in each pixel, with high values indicating edges or sharp intensity changes. Gx for horizontal direction (x-axis) and Gy for vertical direction (y-axis). If the magnitude value (E) is large, it indicates the presence of significant edges or boundaries in the image. In Figure 4, the results of image segmentation using sobel features on the dataset are shown.

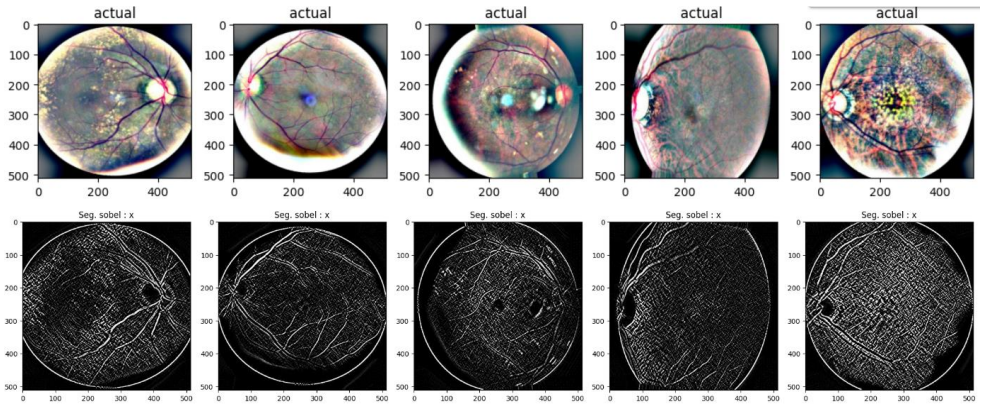

Fig. 5. Visualization of Sobel Feature Segmentation Results

After the segmentation process, the next step is feature extraction using three methods. The first feature extraction uses the HuMomen feature. This feature is a set of seven invariant moments used to capture shape-related features in images [21],[22]. These moments reflect the geometric characteristics of an object in the image, such as roundness, angles, and symmetry. In this study, HuMomen is extracted from the segmented image. It is useful for describing the shape features of structures such as blood vessels or lesions in the retina [23]. This feature is invariant to translation, rotation, and scaling, making it suitable for use in shape recognition applications. This makes it very useful in describing the shape or structure of an object regardless of orientation or scale.

The second feature extraction used is LBP. LBP is a method used to calculate texture and gradient features in images. It describes how the intensity of pixels in an image region changes, and then encodes it in the form of a binary pattern. LBP is used to delineate regions with rough or smooth textures in retinal images. This information can be useful in identifying structures such as blood vessels that have distinctive textures [24]. LBP works by measuring texture patterns at the pixel level by comparing the intensity value of a pixel with that of its neighboring pixels. If the intensity of a neighboring pixel is higher or equal to the center pixel, the neighbor is labeled 1, otherwise, it is labeled 0. Then, these labels are assembled into a binary pattern.

The last feature extraction uses Haralick Texture Features. This method is a set of statistics used to describe the texture characteristics in an image. It includes parameters such as contrast, correlation, energy, and homogeneity, which measure intensity variations and patterns in the image [25],[26]. Haralick's texture features are derived from the Gray Coincidence Matrix (GLCM). GLCM measures the relative frequency of pairs of pixels of a certain intensity located at a certain distance and orientation from each other in the image [27]. Haralick texture features are extracted from the segmented image and help in identifying texture patterns in the retina [28],[29]. From this matrix, various texture metrics, such as contrast, correlation, energy, and homogeneity, can be calculated to characterize the texture of the image. This helps to identify areas with significant differences in texture, which can be an indicator of certain structures or anomalies.

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2.2.3. Algorithm

Decision Tree is a machine learning algorithm that models decisions as a tree with nodes representing features and branches representing decisions based on those features. The tree is built by selecting features that maximize a certain separation criterion, such as Information Gain or Gini Impurity, at each node [1],[30]. This process continues recursively until it reaches a leaf node or meets certain stopping criteria, resulting in a tree structure that is easy to interpret for classification or regression [31],[32]. Decision Tree allows to create decision rules based on the extracted features [33], making it possible to classify these images into appropriate classes. Figure 6 shows an illustration of the decision tree algorithm.

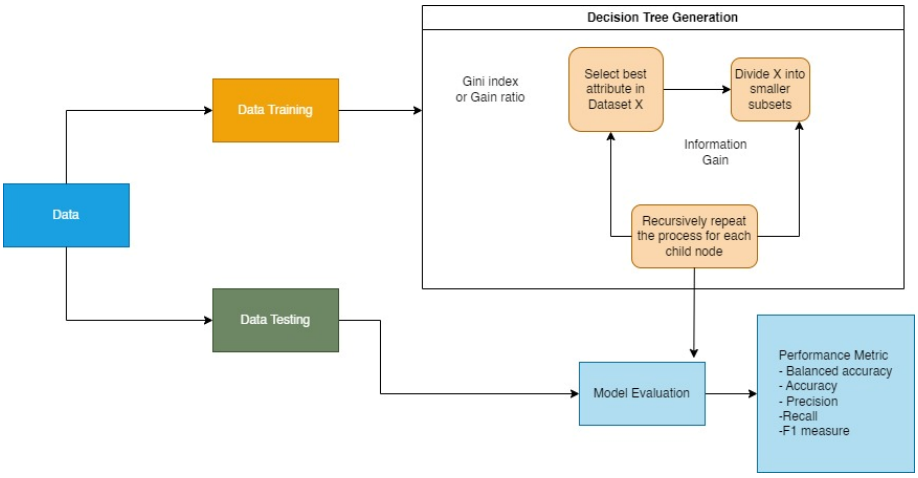

Fig. 6. Illustrating Decision Tree

2.2.4. Performance Analysis

The performance of the Decision Tree classification model will be evaluated using the metrics obtained from each of the three paired segmentation and feature extraction combinations. The purpose of this analysis is to understand the extent to which each combination of segmentation and feature extraction contributes to the model's ability to classify retinal images into the appropriate classes. These metrics include balanced accuracy, accuracy, precision, recall, and f-measure. The equations of each of these metrics are in 4 – 8.

The model evaluation process will also involve the use of 5-fold cross-validation, where the dataset will be divided into five folds, and the model will be trained and evaluated alternately on four folds while one fold is used as the testing set. The dataset division uses an 80-20 ratio, with 80% of the data used for model training and 20% for testing. This cross-validation approach aims to provide more stable and consistent evaluation results, and minimize the impact of dataset variations on model performance.

Balanced accuracy measures the overall accuracy of the model, taking into account the less dominant classes.

$$Balanced\ Accuracy = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right) \tag{4}$$

Accuracy describes the extent to which the model can classify retinal images to measure the ratio of correct predictions to total predictions.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{5}$$

Precision measure the extent to which the images classified as positive by the model are actually positive.

$$Pericision = \frac{TP}{TP+FP} \tag{6}$$

Recall indicate the extent to which the model can detect all true positive cases.

$$Recall = \frac{TP}{TP+FN} \tag{7}$$

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The F-measure value is a combination of Precision and Recall and will provide a more complete understanding of the model's performance.

$$F - measure = \frac{2\, precision \times recall}{precision + recall} \tag{8}$$

The formula above explains:

TP (True Positive) is the number of correct positive predictions.

TN (True Negative) is the number of correct negative predictions.

FP (False Positive) is the number of false positive predictions.

FN (False Negative) is the number of false negative predictions.

The results of the performance analysis will be compared between the three combinations of segmentation and feature extraction. This comparison will help in evaluating the strengths and weaknesses of each approach in classifying various retinal conditions. Thus, in this performance analysis, will evaluate and compare the results of three different approaches in retinal image processing with the aim of understanding how well each of these methods supports the task of retinal condition classification.

3. Results and Discussion

In this section, we present and analyze the results of the comparative performance analysis of various extractions on six classes of retinal datasets. The results are presented in the form of tables and boxplots, to provide insight into the effectiveness of each algorithm. The results include performance comparison of three different feature extraction methods, namely HuMoments, Haralick Texture Features, and Local Binary Pattern (LBP) with three different segmentation techniques namely Canny, Thresholding Function, and Sobel. The performance comparison results can be seen in Table 2 to Table 4 .

Table 2. Performance Comparison Results With Canny Segmentation

∑ Average	HuMoments	Harralick	LBP
Balanced accuracy	0.25	0.29	0.48
Accuracy	0.46	0.49	0.60
Precision weighted	0.35	0.36	0.49
Recall weighted	0.46	0.49	0.60
F1 Weighted	0.33	0.37	0.50

The comprehensive evaluation presented in table 2 provides LBP method has the best performance among the three methods with the highest balanced accuracy, accuracy, precision weighted, recall weighted, and F1 weighted values. This suggests that LBP may be more effective in extracting features from retinal images in the context of this test.

Meanwhile, the HuMoments and Haralick methods, while contributing to the analysis, may require further parameter adjustments or optimizations to improve their performance. The implication is that careful selection of feature extraction methods has a significant impact on the model's ability to recognize patterns and understand the complexity of the retinal image dataset under study.

The balanced accuracy for all feature extractions reflects the extent to which the model is able to classify equally between positive and negative classes. The accuracy value indicates how well the model can correctly classify all samples. Weighted precision provides information on the extent to which the model can accurately identify the positive class. Weighted recall measures the extent to which the model can correctly detect all instances of the positive class. Finally, the weighted F1 value calculates the harmonic mean of the weighted precision and recall. The overall metric shows that LBP is superior to HuMoment and Haralick.These findings contribute crucial insights to the overarching research goal of enhancing the accuracy of retinal disease diagnosis, providing a foundation for more informed medical decision-making.

Table 3. Function Performance Comparison Results With Treshold Function Segmentation

∑ Average	HuMoments	Harralick	LBP
Balanced accuracy	0.21	0.46	0.52
Accuracy	0.43	0.60	0.62
Precision weighted	0.25	0.44	0.53
Recall weighted	0.43	0.60	0.62
F1 Weighted	0.29	0.47	0.53

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The findings delineated in table 3 furnish a detailed exposition on the performance metrics ensuing from the application of the Treshold Function segmentation. The LBP method produced higher scores on all evaluation metrics, including balanced accuracy, accuracy, precision weighted, recall weighted, and F1 weighted, compared to the HuMoments and Haralick methods. A high balanced accuracy value indicates that the model can provide good performance for all classes. A high accuracy value indicates that the model can perform well for all classes, including the dominant class. A high weighted precision value indicates that the model can provide good prediction accuracy for less dominant classes. A high weighted recall value indicates that the model can detect all positive data, including the less dominant positive data. A high weighted f1 value indicates that the model can provide good prediction accuracy and correct positive prediction ratio.

This result indicates that the treshold function approach in segmentation may be a more effective choice in classifying the variations in this test case. This conclusion suggests that LBP, when applied to the treshold function segmentation process, may be more effective in extracting relevant features from retinal images and improving the model's ability to classify such images.

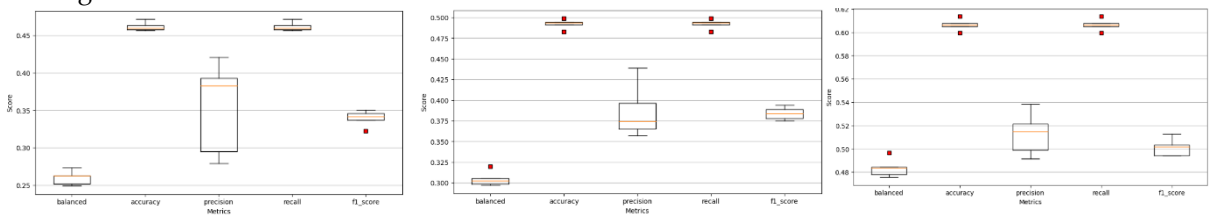
Table 4. Performance Comparison Results With Sobel Segmentation

∑ **Average**	**HuMoments**	**Harralick**	**LBP**
Balanced accuracy	0.26	0.38	0.48
Accuracy	0.46	0.54	0.62
Precision weighted	0.30	0.44	0.58
Recall weighted	0.46	0.54	0.62
F1 Weighted	0.32	0.45	0.55

Table 4 presents a meticulous examination of the model's performance subsequent to Sobel segmentation. The metric of balanced accuracy offers a discerning perspective on the feature extraction methods' efficacy in realizing consistent classification across diverse retinal conditions. The results show that LBP method showed the highest performance with a balanced accuracy value of 0.48, indicating a good level of agreement in classifying between positive and negative classes in retinal images that have undergone Sobel segmentation.

It can be concluded that LBP shows superior performance in most evaluation metrics, including balanced accuracy, accuracy, precision weighted, recall weighted, and F1 weighted. Haralick, although providing good performance, has a slightly lower value compared to LBP. Texture information is important for medical condition detection in retinal images, as many medical conditions cause texture changes in the retina.

In addition to presenting the results of the analysis in tabular form, visualization is also given on the distribution of the values of each metric. To gain deeper insights into the performance of different feature extraction methods for retinal fundus image classification under various segmentation techniques, it is have not only conducted quantitative analyses but also employed a visual approach. This involved creating candle stick visualisation for each model test on each image segmentation, allowing for a more nuanced understanding of the data and the interplay between feature extraction and segmentation.



(a) (b) (c)

Fig. 7. Comparison Results of Performance Test Visualization with Canny Segmetation: (a) HuMoment; (b) Haralick (c) Local Binary Pattern

In Figure 7 for (a), it can be seen that the balanced accuracy has a range (IQR) of the first quartile to the third quartile is between 0.249 to 0.273. While the median value (center line of the box) is about 0.262. For the accuracy value range of the first quartile to the third quartile is between 0.457 to 0.463 and the median is about 0.459. This value indicates that there is not too much variation. In addition, analysis

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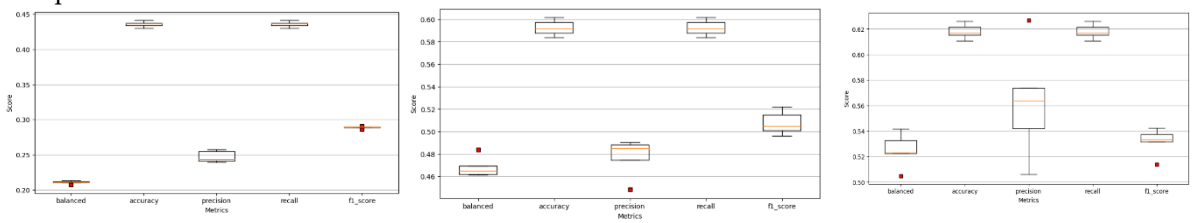
of the Precision and F1-Score metrics also revealed low values outside the lower quartile. This suggests that the model may have difficulty in correctly identifying certain classes. However, the Accuracy and Recall metrics showed relatively smaller variations, indicating good consistency in the model's performance on these criteria.

In figure for (b) all metrics show moderate and fairly consistent model performance. The inter-quartile range for each metric is relatively narrow, indicating that model performance does not fluctuate drastically. The average balanced accuracy is 0.3, with a range of 0.297-0.319, indicating that the model as a whole can achieve a decent balance between detecting positive and negative classes. For the average accuracy value of 0.49, with a range of 0.482-0.499, indicating that the model on average can classify the examples correctly. The average precision metric is 0.38, with a range of 0.357-0.439. The average recall value is 0.49, the same as accuracy, indicating that the model can identify all positive examples. Finally, the average value of F1-score was 0.38, with a range of 0.374-0.394, indicating that the model achieved a moderate balance between precision and recall.

While in figure for (c), compared to the previous results, all metrics show significant improvement. The mean value and inter-quartile range are higher, indicating better model performance. In the balanced quartile range accuracy, recall, and accuracy still have the highest average values. This shows that the model still maintains a good balance between detecting positive and negative classes, while improving its overall accuracy. Precision and F1-score, which were previously lower, saw a more noticeable improvement. This shows that the model has reduced its bias towards predicting all instances as positive and has become more effective in accurately identifying positive examples.



(a) (b) (c)

Fig. 8. Comparison Results of Performance Test Visualization Tresholding Segmetation: (a) HuMoment; (b) Haralick (c) Local Binary Pattern

Overall in Figure 8 for the categories of figures (a) shows good consistency in classifying positive and negative classes, as reflected by the relatively stable balanced accuracy value and low variation in other metrics. There is little variation in the distribution of balanced accuracy values, with a range (IQR) of 0.211 to 0.212. The accuracy of the model remains at a consistent level, providing correct predictions in general. Precision indicates the model's ability to provide correct positive predictions with a good level of consistency. The model's performance in detecting positive classes (recall) also appears stable. Precision and F1-score experienced a significant decrease. Both of these metrics, which measure the accuracy of predicting the positive class, showed a decrease compared to the previous analysis. This means that the model tends to misclassify many examples, although it still maintains overall balance and accuracy.

In addition, figure 8 (b) also shows that mproved overall performance compared to before. The mean value and inter-quartile range are higher, indicating better model performance. It can be concluded that the model has a good ability to classify classes in a balanced manner, as seen from the stable balanced accuracy and high recall values. Although accuracy is relatively stable, the variation between measurements can reach a considerable range, indicating some fluctuations in model performance. Precision, which measures the ability of the model to give correct positive predictions. The F1-score illustrates the balance achieves a good balance between precision and the ability to detect positive classes.

Figure 8 c specifically indicates that the balanced accuracy and stable recall values indicate the ability of the model to classify classes in a balanced manner. The accuracy of the model is relatively high

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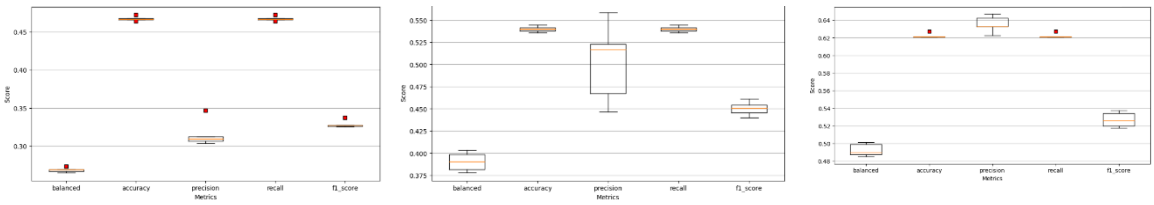
and stable, indicating that the model is able to provide correct predictions overall. On the precision side, the model shows a good ability to give positive predictions, and the F1-score illustrates an adequate balance between precision and recall. The variation in the distribution of the values of the evaluation metrics, such as accuracy and f1-score, is quite small, indicating the stability of the model's performance in this test.



(a) (b) (c)

Fig. 9. Comparison Results of Performance Test Visualization Sobel Segmetation: (a) HuMoment; (b) Haralick (c) Local Binary Pattern

Figure 9 (a) shows that sufficient ability to classify classes in a balanced manner. There is variation in the distribution of balanced accuracy values, with a range (IQR) of 0.269 to 0.273. The variation in the distribution of accuracy values is quite small, with an (IQR) range between 0.466 to 0.472. Meanwhile, the distribution of precision values has a range (IQR) between 0.304 to 0.347. The variation in the distribution of recall values is quite small, with a range (IQR) between 0.466 to 0.472. The distribution value of the f1-score value is in the range between 0.325 to 0.337.

Then in figure (b) the box plot shows that Variation in the distribution of balanced accuracy values, with a range (IQR) between 0.378 to 0.403. The accuracy values are quite small, with a range (IQR) between 0.536 to 0.542. For the distribution of precision values, with a range (IQR) between 0.468 to 0.523. The recall value is quite small, with a range (IQR) between 0.536 to 0.545. The f1-score value also shows variation with a range (IQR) between 0.441 to 0.461.

Whereas in figure (c) all metrics show significant improvement compared to the last test results with Sobel Segmentation and LBP alone. The mean values and inter-quartile range are higher, indicating better model performance. Although the recall did not experience a large increase, the value remained stable and moderate. This shows that the model maintains the level of ability to detect positive instances. As for the F1-score, although it increased, it is still below precision. This indicates that recall, although stable, remains a point that needs to be improved for a better balance between precision and recall.

The results obtained from the comparative analysis of different segmentation techniques and feature extraction methods on retinal fundus images provide valuable insights into the effectiveness of these approaches. Notably, the Local Binary Pattern (LBP) method consistently outperformed HuMoments and Haralick across various segmentation strategies, as indicated by higher scores in balanced accuracy, accuracy, precision, recall, and F1 weighted. This suggests that LBP captures more discriminative features for distinguishing between different retinal conditions. The superior performance of LBP suggests its efficacy in capturing intricate details related to shape, gradient, and texture in retinal images, essential for accurate disease classification.

This findings align with some existing research that highlights the effectiveness of LBP for retinal image analysis [34] demonstrate the successful use of LBP for optic disc detection, while [35] utilize LBP in their multi-modal retinal blood vessel segmentation approach. However, research extends these findings by comparing LBP with other feature extraction methods and analyzing its performance across different segmentation techniques. Additionally, this research contributes to the growing body of literature on retinal image analysis using large-scale datasets and benchmark datasets, as advocated [36]. This research complements these efforts by showcasing the impact of segmentation and feature extraction on the performance of a Decision Tree classifier, providing a holistic view of the image analysis process.

The improved accuracy and generalizability of LBP-based models have numerous practical implications for retinal disease diagnosis. Firstly, it can lead to the development of more accurate and

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reliable automated diagnostic tools, potentially aiding in early detection and treatment of retinal diseases. Secondly, LBP's robustness to different segmentation techniques makes it suitable for real-world applications where image quality can vary. This can improve the accessibility and affordability of retinal disease diagnosis, particularly in resource-limited settings.

Research has limitations that need to be addressed in future studies. Firstly, the study focused on a specific set of retinal diseases. The dataset used, while comprehensive, may not encompass the full spectrum of retinal conditions. Further research is needed to evaluate the generalizability of LBP to a wider range of retinal pathologies. Secondly, while we explored three common segmentation techniques, investigating the performance of LBP with more advanced segmentation algorithms could provide further insights. Additionally, the research did not delve into the interpretability of LBP features, which could be valuable for understanding how the model makes its predictions.

To further develop this research, future studies could explore several exciting avenues. Building upon the promising results of our research, future work can explore several exciting avenues. Firstly, expanding LBP's application to encompass a wider range of retinal diseases and pathologies would serve to validate its generalizability and solidify its place as a robust feature extractor. Secondly, evaluating LBP's performance when combined with other advanced feature extraction and segmentation techniques could unlock further accuracy gains and reveal synergistic potential. Thirdly, developing interpretable LBP-based models would offer crucial insights into the model's decision-making process, allowing for bias identification and mitigation. Lastly investigating the combination of deep learning techniques with traditional methods could also be valuable. By diligently addressing the limitations of current study, can refine LBP's effectiveness and contribute to the development of more accurate and accessible diagnostic tools for retinal diseases, ultimately improving patient outcomes and advancing the field of ophthalmic medicine.

4. Conclusion

Conclusions written in one paragraph, presented briefly, narrative, non-bulleted, and conceptual. The research impact must be state In this series of studies conclusively demonstrated that Local Binary Pattern (LBP) reigns supreme as the feature extraction champion for retinal fundus image classification, outperforming both HuMoments and Haralick Texture Features across all segmentation techniques. LBP consistently delivered superior performance in accuracy, precision, recall, and F1-score, proving its prowess in capturing discriminative features that distinguish diverse retinal conditions. Remarkably, LBP maintained its dominance even when paired with different segmentation methods, showcasing its adaptability and generalizability to diverse image characteristics. While a closer look at boxplots revealed LBP's consistent performance in accuracy and F1-score, variations in precision and recall warrant further investigation. This research offers not only a clear answer to initial hypothesis, but also valuable insights for future endeavors. This research goes beyond conventional approaches by emphasizing the practical implications of these findings, fostering a deeper understanding of the intricate details in retinal images that are critical for accurate disease classification. By offering a comparative perspective, this study helps in improving the methodology for automated retinal disease diagnosis.

By establishing LBP as the preferred feature extraction method and highlighting the importance of segmentation technique choice, pave the way for developing more accurate and reliable retinal disease diagnostic tools. Furthermore, findings prompt further exploration of LBP's variability and potential in combination with other advanced techniques, ultimately aiming to refine its effectiveness and unlock new applications across the medical image classification landscape. This research is just the beginning, and the exciting future holds the promise of improved patient care and advancements in ophthalmic medicine thanks to the power of LBP.

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Author Contributions

F. Author: Conceptualization, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, and writing - review & editing. S. Author: Data curation, investigation, resources, software, validation, visualization, and writing - original draft. T. Author: Investigation, visualization, and writing - review & editing. F. Author: Investigation, project administration, and writing - review & editing. (Author Contribution Form All authors must suitable with the relevant terms to indicate their contributions. To know more about the CReDiT Author Statement and definitions of each term mentioned, please visit https://casrai.org/credit/.)

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Declaration of Competing Interest

We declare that we have no conflict of interest.

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